

PHOTONICS Research

Latent-wavefront Fourier ptychography for stained tissue microscopy

SHUHE ZHANG,¹ JIAYUN LI,² AND LIANGCAI CAO^{1,*} 

¹Department of Precision Instruments, Tsinghua University, Beijing 100084, China

²St. Anne's College, University of Oxford, Oxford OX2 6HS, UK

*Corresponding author: clc@tsinghua.edu.cn

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Fourier ptychographic microscopy (FPM) is a promising technique for achieving high-resolution and large field-of-view imaging, which is particularly suitable for pathological applications, such as imaging hematoxylin and eosin (H&E) stained tissues with high space-bandwidth and reduced artifacts. However, current FPM implementations require either precise system calibration and high-quality raw data, or significant computational loads due to iterative algorithms, which limits the practicality of FPM in routine pathological examinations. In this work, latent wavefront denoting the unobservable exiting wave at the surface of the sensor is introduced. A latent wavefront physical model optimized with variational expectation maximization (VEM) is proposed to tackle the inverse problem of FPM. The VEM-FPM alternates between solving a non-convex optimization problem as the main task for the latent wavefront in the spatial domain and merging together their Fourier spectrum in the Fourier plane as an intermediate product by solving a convex closed-formed Fourier space optimization. The VEM-FPM approach enables a stitching-free, full-field reconstruction for Fourier ptychography over a 5.3 mm × 5.3 mm field of view, using a 2.5× objective with a numerical aperture (NA) of 0.08. The synthetic aperture achieves a resolution equivalent to 0.53 NA at 532 nm wavelength. The execution speed of VEM-FPM is twice as fast as that of state-of-the-art feature-domain methods while maintaining comparable reconstruction quality. © 2025 Chinese Laser Press

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1. INTRODUCTION

The pathology laboratories market has been driven by the increase in the number of healthcare facilities in developing nations, the growing demand for routine medical check-ups, and the enhanced reimbursement policies for diagnostic tests. In 2021, the global market size of pathology laboratories was valued at USD 311.2 billion and is expected to expand at a compound annual growth rate of 7.86% from 2022 to 2030 [1]. Within, bright-field whole-slide microscopy plays a crucial role in pathological analysis with H&E stained samples [2,3]. Bright-field whole-slide microscopy enables pathologists to examine entire tissue slides in great detail [4]. This comprehensive visualization helps identify and analyze cellular and tissue structures, which is essential for accurate diagnosis. With the integration of artificial intelligence and machine learning algorithms [5], bright-field whole-slide microscopy facilitates automated analysis [6], pattern recognition [7], and predictive modeling [8], further enhancing diagnostic efficiency and uncovering insights that might be missed by human evaluation alone. Fourier ptychographic microscopy (FPM) offers a potential solution for large-FOV (field of view), high-resolution imaging of H&E stained samples [9]. The FPM combines

programmable illumination and synthetic aperture, providing a cunning solution to bypass the trade-off between large FOV and high optical resolution. The in-born advantages of FPM make it a promising solution for visualizing H&E stained samples and performing related pathological analyses including tissue segmentation and cell cluster analysis [10]. In addition, the intermediate outputs of FPM, such as phase information [11,12], system aberration [13,14], and diffraction tomography [15–17], extend the scope of FPM-related applications. Compared to bright-field whole-slide microscopy, FPM has great potential not only for bright-field pathological analysis [18] but also for applications like label-free microscopy, 3D tomography [19], and digital adaptive optical microscopy [20].

However, transitioning to FPM for pathological applications introduces a set of challenges. Uncertainties such as illumination direction [21], noise in dark-field images, and inconsistencies in illumination intensity are inevitable and cannot be completely eliminated [22]. Traditional FPM reconstructions are sensitive to system uncertainties and rely heavily on data-clearing procedures [21,23]. Developing a robust reconstruction framework to address system uncertainties is crucial for practical FPM applications in the field of pathology.

One approach to eliminate uncertainty is to parameterize uncertainties within the forward model (image formation model) of FPM and optimize them by calculating the gradient [24] or using auto-differentiation techniques. However, gradient-based reconstruction significantly increases the computational complexity, rendering them impractical for large-scale FPM reconstructions. Furthermore, due to the non-convex nature of the FPM inverse problem, finding an optimal solution remains challenging [25–27]. The recently developed feature-domain FPM (FD-FPM) addresses this issue by formulating the loss function based on image features [28,29], reducing the impact of the vignetting effect [30], LED position misalignment, and noise [31]. However, FD-FPM requires a long reconstruction duration due to its non-convex nature and involves frequent Fourier and inverse Fourier transforms during each iteration. Therefore, there is a pressing need for new computational frameworks that can balance fast execution speeds with high-quality reconstruction, especially for pathological applications of FPM.

Revisiting the conventional FPM and related synthetic aperture techniques, the complex wavefront of the optical waves at the surface of the optical sensor, denoted as $\mathbf{Z}$, plays a crucial role in the successful reconstruction of the sample in FPM. Each $\mathbf{Z}$ corresponds to a sub-aperture of the sample's Fourier spectrum. If $\mathbf{Z}$ were known in advance, such as through interference measurements, the sample could be reconstructed simply by stitching together the individual $\mathbf{Z}$ to form the complete Fourier spectrum for aperture synthesis. However, due to the loss of phase information when optical waves are detected by sensors, the inverse problem in FPM has been traditionally treated as a phase retrieval problem [32,33]. This phase retrieval problem is solved by iteratively applying amplitude constraints to $\mathbf{Z}$ and Fourier space constraints to the sample's wavefront using the ptychographic iterative engine (PIE) and its variations. Despite this, the treatment of $\mathbf{Z}$ has not been fully optimized for FPM as $\mathbf{Z}$ is considered to be an intermediate product during the phase retrieval. In contrast, the FPM inverse problem can be framed as a regression task involving unobserved/latent variables $\mathbf{Z}$, as seen from the perspective of machine learning [34–36]. Given the presence of latent variables, the inverse problem in FPM can be naturally addressed within the expectation-maximization (EM) framework where $\mathbf{Z}$ is treated as the main target to be optimized but has hitherto been overlooked in the FPM community.

In this work, we propose the latent wavefront physical model optimized with variational expectation maximization (VEM) to tackle the inverse problem of FPM for H&E tissue imaging over a large FOV. The VEM-FPM alternates between solving a non-convex optimization problem as the main task for the latent wavefront, $\mathbf{Z}$, in the spatial domain, and merging together their Fourier spectrum in the Fourier plane as an intermediate product by solving a convex closed-formed Fourier space optimization. This approach represents the first probabilistic model with latent variables for solving the FPM inverse problem. Unlike gradient descent-based methods [28,37], the VEM-FPM provides a semi-closed-form solution for FPM reconstruction. Additionally, VEM-FPM requires no auxiliary

variables, making it more computationally efficient than ADMM methods [38]. The VEM-FPM method converges rapidly and imposes a lower computational load, particularly for large FOV FPM reconstruction, as it avoids the need to compute the gradients of the loss function with respect to the large-scale sample wavefront. The framework of VEM-FPM will be detailed in Section 2 followed by some results on both bright-field whole-slide FPM reconstruction and quantitative phase imaging.

2. FRAMEWORK OF VEM-FPM

In the FPM imaging system as shown in Fig. 1(a), the forward model of FPM imaging can be separated into two parts. (1) First the sample's wave field is propagated through the imaging system. It arrives at the camera sensor plane, which is given as $\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_n \in \mathbf{Z}$, with $\mathbf{Z}_n = \mathbf{F}^\dagger \mathbf{P} \mathbf{M}_n \Phi$, where $\mathbf{M}_n$ denotes the selection matrix for the n th LED illumination, $\mathbf{P}$ represents the pupil function, and $\mathbf{F}^\dagger$ stands for the inverse Fourier transform. † denotes the Hermitian transpose of a matrix. $\mathbf{Z}$ is the unobserved (latent) wave field arriving at the camera's sensor plane. Since the camera sensor can only capture the intensity of the optical wavefront, the (2) second part of the forward model is given as $I_n = |\mathbf{Z}_n|^2 + \eta$, where η denotes noise signals. Based on the two-step image formation model, $\mathbf{Z}$ is regarded as a latent wavefront that connects the complex-domain wavefront with real-domain intensity measurements. Through regression, we aim to recover the Fourier spectrum of the sample's wavefront, Φ , together with the pupil function $\mathbf{P}$ using a set of observed intensity measurements $I_1, I_2, \dots, I_n \in \mathbf{I}$.

However, due to the difficulty in directly obtaining the probability of the latent variable $\mathbf{Z}$, we use the VEM algorithm to maximize the likelihood function. The VEM-FPM solves the regressive iteratively by obtaining a proximate estimation of probability for the latent $\mathbf{Z}$ during the E -step, and then maximizing the expectation in the M -step as shown in Fig. 1(b). During the M -step, the prior distribution of parameters is taken into consideration to provide regularization. At the t th iteration, the VEM-FPM performs two steps including the E -step and the M -step, which are given as follows.

- E -step. Updating latent variable $\mathbf{Z}_n$:

$$\mathbf{Z}_n = \mathbf{X} + \eta \cdot \text{Diag}[\nabla^\top \text{sign}(\nabla \sqrt{I_n} - \nabla |\mathbf{X}|)] \frac{\mathbf{X}}{|\mathbf{X}|} \Big|_{\mathbf{X}=\mathbf{F}^\dagger \mathbf{P}^{t-1} \mathbf{M}_n \Phi^{t-1}}. \quad (1)$$

- M -step. Updating Φ and $\mathbf{P}$ using $\mathbf{Z}_n$:

$$\Phi^t = \frac{\nabla^\top \nabla \sum_{n=1}^N \mathbf{M}_n^\top (\mathbf{P}^{t-1})^\dagger \mathbf{F} \mathbf{Z}_n + \alpha \mathbf{F} \hat{\mathbf{U}}}{\nabla^\top \nabla \sum_{n=1}^N \mathbf{M}_n^\top |\mathbf{P}^{t-1}|^2 \mathbf{M}_n + \alpha}, \quad (2)$$

$$\mathbf{P}^t = \frac{\nabla^\top \nabla \sum_{n=1}^N (\mathbf{M}_n \Phi^{t-1})^\dagger \mathbf{F} \mathbf{Z}_n + \beta \hat{\mathbf{P}}}{\nabla^\top \nabla \sum_{n=1}^N |\mathbf{M}_n \Phi^{t-1}|^2 + \beta}, \quad (3)$$

where $\hat{\mathbf{U}}$ denotes the post-processed (denoising) output of the sample's wavefront in the spatial domain. $\hat{\mathbf{P}}$ denotes the post-processed (denoising) output of the pupil function. Penalty parameters α and β increase w.r.t. iterations. The flowchart of VEM-FPM is plotted in Fig. 1(c). Equation (2) denotes

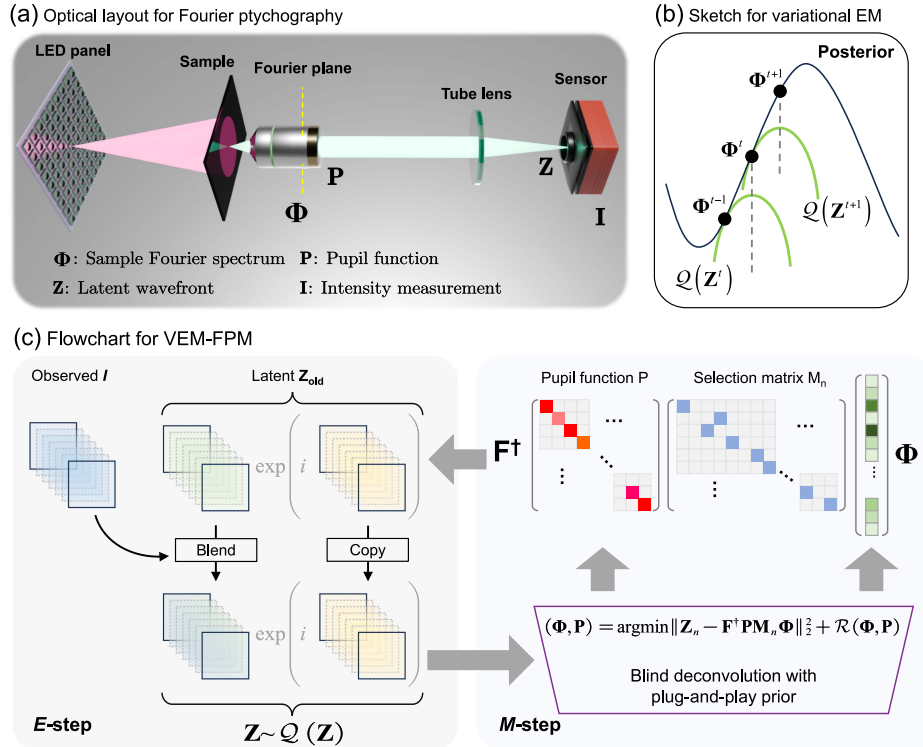


Fig. 1. Sketch for VEM-FPM. (a) Optical layout of a Fourier ptychographic microscopy system. (b) Sketch for the variation EM algorithm. The VEM alternatively finds the estimation of expectation denoted by the green parabola (*E*-step) and maximizes the likelihood by locating the lateral coordinate of the vertex of the parabola (*M*-step). (c) Flowchart of VEM-FPM. The latent variable Z is estimated in the *E*-step through the observed data and predicted data generated by the forward model. In the *M*-step, the sample wavefront Φ and the pupil function P are obtained through blind deconvolution by solving the quadratic loss function with the plug-and-play prior for in-loop image denoising.

the deconvolution in Fourier space that stitches together the Fourier spectrum of each sub-aperture corresponding to individual latent wavefront in the sample's Fourier plane. Equation (3) denotes the deconvolution for solving the pupil function. Laplacian ($\nabla^T \nabla$), deduced from the retinex theory [39], removes the slow-varying uneven illuminations and maintains the intensity consistency across the entire image. Equations (2) and (3) have similar forms to that of different map algorithms in early implementation of ptychography [40] as the noise signal in Fourier space is assumed to follow Gaussian distribution. By assuming different noise distributions, alternative solutions for *M*-step can be deduced, but the execution speed is slower than the closed-form solution in Eqs. (2) and (3). Please refer to Ref. [41] for detailed information.

Unlike gradient-based methods where the loss function is non-convex and optimized using gradients and backpropagation, the VEM-FPM approach decomposes the loss function into two components for optimization in the *E*-step and *M*-step. In the *E*-step, the loss function is non-convex and is minimized using proximal gradient descent. The goal of the *E*-step is to estimate the expectation of the latent variable Z , which represents the wavefront of the exit pupil arriving at the camera sensor's surface. Once Z is estimated, a closed-form deconvolution can be applied to efficiently solve for parameters Φ and P , as the loss function in the *M*-step is quadratic. This combination

of proximal gradient descent and quadratic deconvolution accelerates the FPM reconstruction, with the *E*-step computations being fully parallelizable.

In the following contents, all experiments are implemented in a self-established FPM platform with a commercial LED panel from Adafruit. The LED array was controlled by a micro-control unit (Arduino UNO R3, USA), triggered by signals in MATLAB using serial port communication. The intensity of each LED was toggled by a serial port input value ranging from 0 to 255. The red channel has a wavelength of 628 nm, the green channel 532 nm, and the blue channel 470 nm. Codes are composed and implemented using MATLAB R2023b running on a desktop with a 12th Gen Intel(R) Core(TM) i5-12400F 2.5 GHz, 32.0 GB RAM, and an NVIDIA GeForce RTX 3070 graphics card.

3. RESULTS

A. Stitching-Free Reconstruction

The stitching-free FPM reconstruction result for an H&E stained slice (rat colon tissue) is shown in Fig. 2. The ring-shaped programmable LED array is placed 90 mm above the slide to provide angular-varying illumination. The LED array consists of 6 rings with 1, 8, 12, 16, 24, and 32 LEDs embedded, and the radius of each ring increases by 9 mm.

We use an apochromatic 4 $\times$, NA = 0.1 objective lens and a 16-bit camera (pco.edge 4.2M, 6.5 μm , Germany) to capture the images. A total of 93 images were collected for each color channel, which took about 10 s for data collection. Raw images for the red channel are shown in the left top of Fig. 2(a), where the vignetting effect appears due to the large FOV and nonlinear effect of the 4f image system. Images for the red, green, and blue channels are separately captured and reconstructed, and then registered and combined to produce a full-color image.

The performance of VEM-FPM is compared against conventional ePIE [13,43] and the FD-FPM methods. Both ePIE and FD-FPM are implemented using default hyperparameters. $\eta = 0.005$ for the VEM-FPM, and a median filter with a 3×3 pixel kernel is applied to the VEM-FPM as the denoiser. Accordingly, both FD-FPM and VEM-FPM achieve vignetting-free reconstruction across the entire FOV. The quantitative intensity profiles, which are the mean value of the RGB color channel at the pixels, are plotted in Fig. 2(b), denoting

the enhancement of resolutions. The ePIE fails to obtain good reconstruction results due to the vignetting effect, which breaks the linear-invariant properties of the FPM's forward model. Compared to the FD-FPM, the VEM-FPM achieves direct and non-blocked reconstruction of the entire FOV without vignetting artifacts. Furthermore, color differences between image segments and the consequent stitching artifacts are efficiently avoided. The retinex-based deconvolution and the plug-and-play denoiser further increase the visual quality of the outputs, enabling noise-suppressed, contrast-enhanced microscopy for the H&E stained sample. The reconstruction quality of VEM-FPM is benchmarked by comparing it with other state-of-the-art (SOTA) methods; please refer to Ref. [41] for details of simulation studies.

B. Large-FOV H&E Microscopy

FPM enables imaging of large areas of tissue while maintaining high resolution, thus reducing the need for multiple imaging

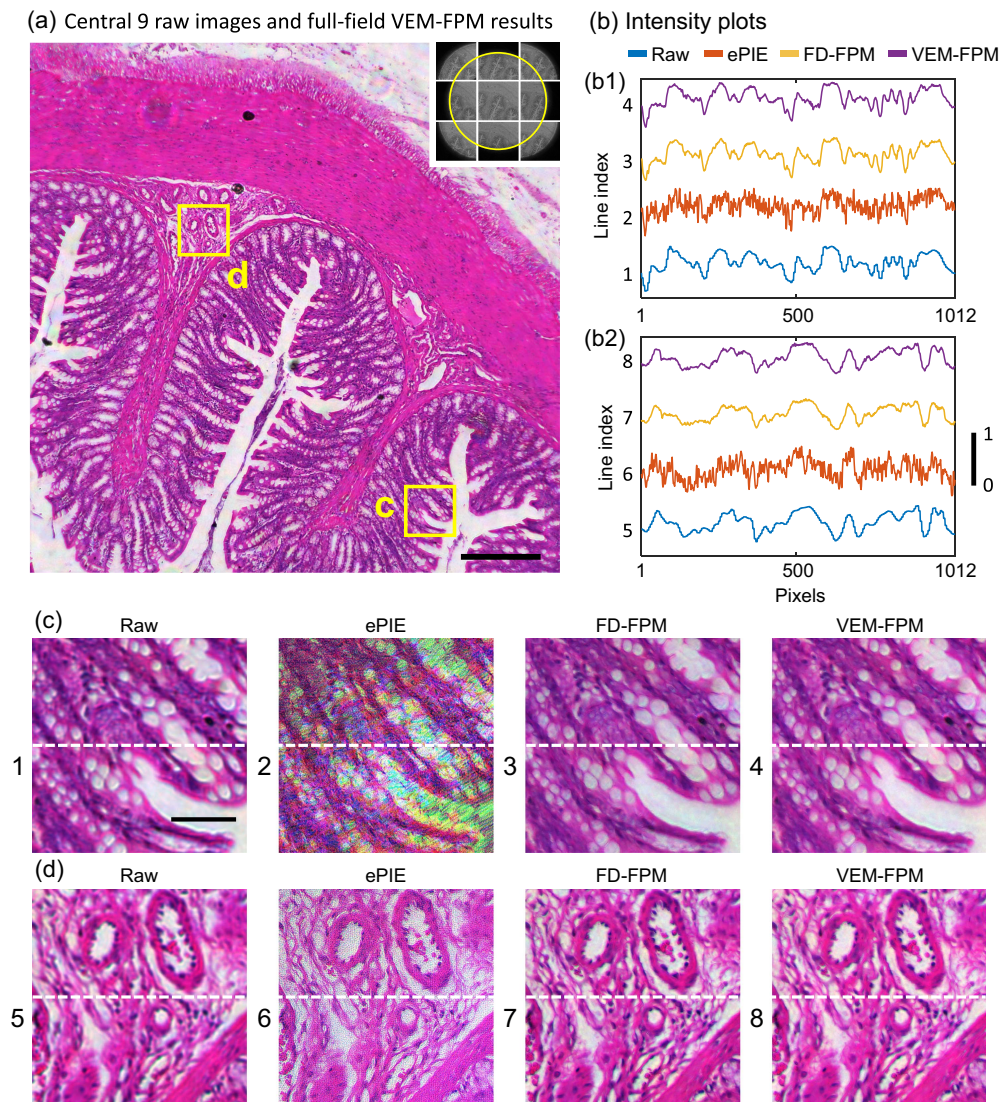


Fig. 2. FPM reconstruction using VEM-FPM. (a) Full-field reconstruction results and the first 9 images for the red channel. (b) Quantitative intensity profile along the white lines in (c) and (d). (c), (d) Zoomed-in image for the yellow boxes in (a). Scale bar: 500 μm for (a); 100 μm for (c) and (d). Full-resolution image is available on Gigapan [42].

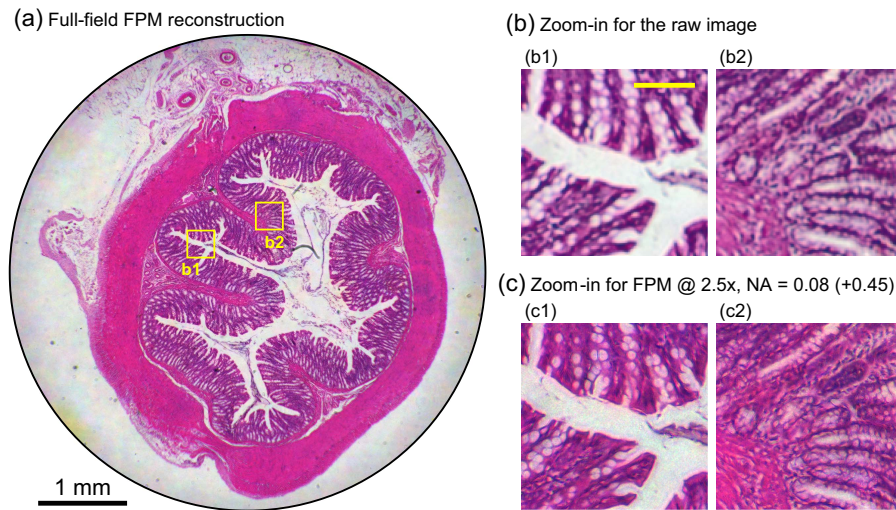


Fig. 3. Large FOV reconstruction using VEM-FPM. (a) VEM-FPM result. (b), (c) Zoomed-in image for yellow boxes in (a). (b1) and (b2) are raw images. (c1) and (c2) are from VEM-FPM. Scale bar: 1 mm for (a); 100 μm for (b) and (c).

scans or tiling, which is particularly useful for H&E stained samples. We apply the VEM-FPM to a large FOV microscope as shown in Fig. 3(a). Here, we use a 2.5 $\times$, NA = 0.08 objective lens, so that the size of the field of view is 5.3 mm $\times$ 5.3 mm. The entire FOV is directly reconstructed, and the output image has 16,384 $\times$ 16,384 pixels. There is no grid-like artifact in the image as shown in Fig. 3(a). Partially zoomed-in images in Figs. 3(b) and 3(c) highlight the increase of optical resolution while maintaining the effect of FOV.

The VEM-FPM overcomes the limitation of FOV for high-resolution microscopy by using a low numerical aperture objective lens to capture a large FOV. Compared with conventional FPM, the VEM-FPM also overcomes the limitation due to the vignetting effect, especially for large FOVs. The low-resolution images acquired under different illumination angles are used to update the latent variables $\mathbf{Z}$ to obtain the sample's complex wavefront at the camera sensor. Through Fourier domain deconvolution, a high-resolution image over the entire FOV is recovered. The properties allow for a large portion of the H&E stained slide to be imaged in one pass while maintaining detailed information at a cellular level. Other experimental studies of VEM-FPM are available in Ref. [41].

C. Quantitative Phase Target

Similar to other FPM solvers, the VEM-FPM is capable of retrieving the phase information of the sample. The experiment is conducted using an apochromatic 4 $\times$, NA = 0.1 objective lens. Figure 4(a) shows the raw image (258 $\times$ 258 pixels) when the central LED is turned on. The reconstructed phase is shown in Fig. 4(b), where the high-frequency components are reconstructed as line structures are recovered for group 9. However, low-frequency components are not fully recovered, which is a common limitation of conventional FPM. Figure 4(c) illustrates the loss function as a function of the number of epochs for VEM-FPM. A learning rate decay was employed to accelerate convergence [31]. All images were used to update the

sample's wavefront within a single epoch, which took approximately 0.09 s, facilitated by the fast deconvolution process of VEM-FPM.

The reconstruction process is notably efficient, as the loss function decreases rapidly within a few epochs. Nevertheless, we observed that the sample's phase was not fully reconstructed. As shown in Fig. 4(e), the line structures for group 6 are gradually recovered as the epoch increases to 700. The changes in phase pattern for group 6 are quantitatively denoted in Fig. 4(f). Phase patterns for group 6 element 6 [at 800 pixels in Fig. 4(f)] are correctly recovered after 400 epochs. Despite the failure in recovering the low frequencies, the high-frequency components are correctly recovered, which can be used to validate the resolution of VEM-FPM. The zoomed-in image in the Fig. 4(d) confirms that element 6 of group 9 (912 lp/mm) was clearly resolved, and even element 1 of group 10 (1024 lp/mm) was recovered, though with weak contrast. Figure 4(g) presents the quantitative phase profile along the white line in Fig. 4(c). The phase profile shows minimal change in phase with respect to the number of epochs, indicating convergence.

When imaging a specimen with any linear phase shift, conventional FPM produces intensity images with constant values across different illumination angles. For any plane wave illumination where the illumination NA within the NA of the objective lens, the phase transfer function (PTF) becomes zero near the origin as the PTF is canceled [44], meaning that phase information at certain low spatial frequencies is permanently lost during data acquisition and cannot be recovered post measurement. Consequently, conventional FPM struggles to achieve accurate quantitative phase imaging, especially for slowly varying phase objects or large, continuous structures. To address this issue, an NA-matching strategy can be employed, ensuring that the illumination NA matches the objective's NA. An alternative method is to integrate coded ptychography by placing a thin coded surface atop the image sensor, as described in Refs. [45,46]. This coded surface converts object phase

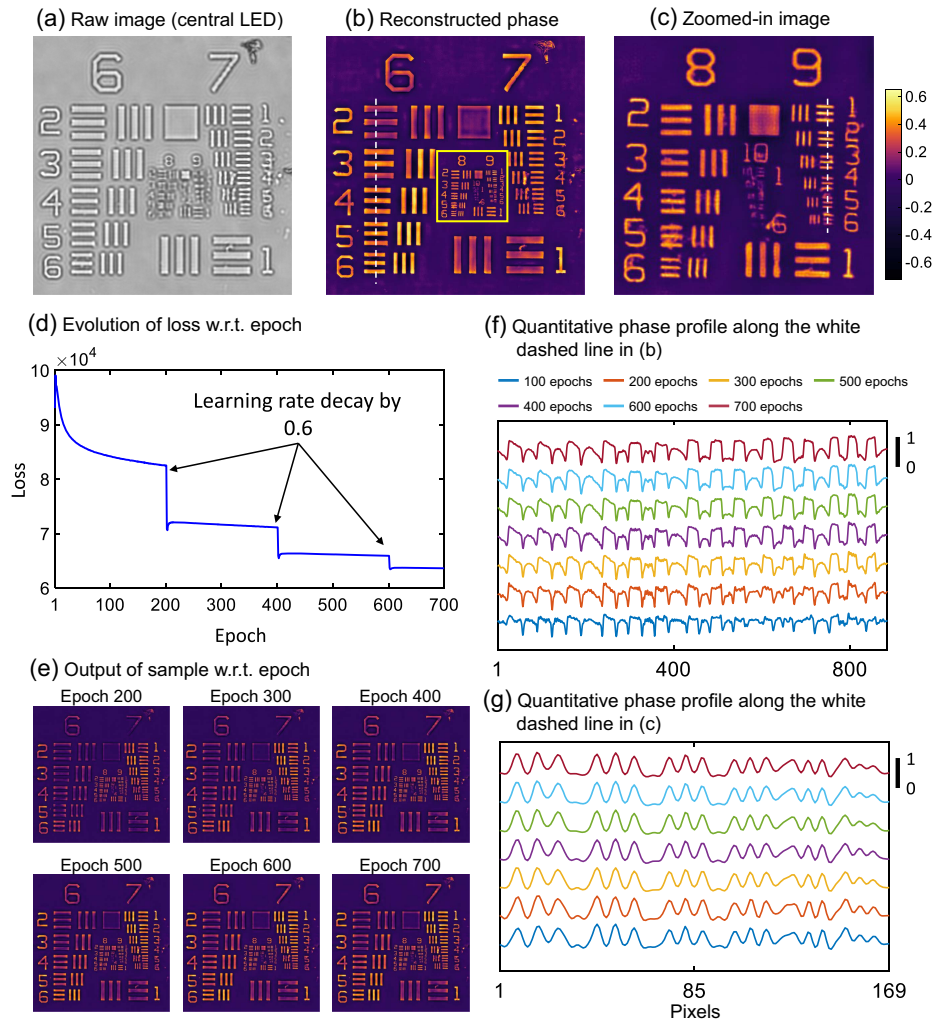


Fig. 4. Quantitative phase reconstruction using VEM-FPM. (a) Raw image. (b) Reconstructed phase pattern. (c) Zoomed-in image denoted by the yellow box in (b). (d) Evolution of the loss function w.r.t. epoch. (e) Evolution of the phase pattern w.r.t. epoch. (f) and (g) Quantitative phase profile along the white lines in (b) and (c), respectively.

information into detectable intensity variations, thereby enabling the recovery of low-frequency phase information that would otherwise be lost in conventional FPM [46].

4. DISCUSSION

Fourier ptychographic microscopy (FPM) offers a combination of wide FOV and high-resolution imaging, making it valuable for a variety of applications, especially for pathological analysis. A wide-FOV and high-resolution image is obtained by solving the regressive problem given by the inverse task of the FPM. Being a regressive task for the FPM inverse problem, the quality of the reconstruction depends critically on how well the forward model can best explain the experimental observation [23,35]. However, applying FPM in practical application is challenging due to the need for an accurate numerical forward imaging model that closely matches real-world imaging systems [47]. The sensitivity makes FPM vulnerable to system misalignment, aberrations, and poor data quality. In conventional FPM, the forward model is deduced from Fourier principles of a 4f

optical system, so that the image process can be simply treated as a linear convolutional process in which the convolutional kernel is provided by the pupil function of the imaging system. However, this linear convolutional process is invalid in practical imaging systems due to spatial-varying aberrations [48,49] and the finite size of the LED's illumination, which introduce vignetting effects.

Spatial-varying aberrations are minimized by the objective lens manufacturers, especially for objectives with low NA. While the use of LED arrays in FPM introduces additional uncertainties due to variation in LED orientations, positions, and intensity distribution [50], the combination of hundreds of LEDs increases the system uncertainty by orders. Replacing the LED with a laser source is a good solution to reduce this uncertainty and vignetting effect [51]. Digital micromirror devices (DMDs) can be employed to precisely control the laser's illumination direction and position on the sample plane. However, incorporating a laser and DMDs increases both the cost and complexity of the FPM system [32]. An alternative strategy is to model these uncertainties within a comprehensive

“super” forward model that not only includes the sample wavefront and pupil function parameters, but also parameterizes the LED positions as learnable variables. By employing optimization techniques such as simulated annealing [52] or gradient descent [24], the LED parameters can be optimized by minimizing the loss function. However, these optimization methods are often inefficient due to the presence of noise and outliers, such as pulse noise and intensity mismatches. Moreover, the non-convex nature of the loss function in phase retrieval further complicates the corrections of uncertainties through optimization. Calculating the gradient of the loss function with respect to LED positions imposes a significant computational burden. For each LED, a large matrix—equivalent in size to the sample wavefront—must be computed and subjected to a Fourier transform. This heavy computation limits the practicality of gradient-based methods in large-scale FPM reconstructions.

The recently developed closed-form FPM (APIC) addresses several theoretical limitations [53]. Under the NA-matching condition, it solves the Fourier spectrum for bright-field components using the Kramers–Kronig (KK) relation [54], while dark-field components are obtained through an interference-like solution. However, the closed-form FPM has practical limitations. One major restriction is the strict NA-matching condition. Additionally, closed-form FPM constrains the loss function to a specific type (L_2 -norm), which is mathematically equivalent to solving a quadratic optimization problem. Consequently, it cannot take advantage of the non-convex nature of regression by allowing for custom loss functions that might better utilize image information. Moreover, the

closed-form FPM cannot effectively address vignetting effect, which demands high-quality raw images for accurate reconstruction.

In the presence of uncertainty in FPM, a wise approach is not necessarily to solve the uncertainty, but to bypass it. Feature-domain FPM (FD-FPM) leverages the image’s feature-domain information to circumvent challenges in the image domain [28,55]. By reformulating the loss function using the image’s first-order gradients (edge features), FD-FPM demonstrates impressive robustness against LED positional misalignment, intensity fluctuations [56], and various noise signals. Although FD-FPM can be accelerated through parallel computation, the batch gradient descent strategy prolongs its convergence time. The frequent Fourier and inverse Fourier transforms significantly increase the computational load, particularly for large field-of-view reconstructions. For example, reconstructing 361 images of 2048×2048 pixels with an upsampling rate of 5 takes approximately 30 min using a 3070 GPU. In contrast, with VEM-FPM, the execution time is reduced to 7 min without sacrificing reconstruction quality. Comparisons between VEM-FPM and APIC, and other SOTA FPM algorithms, are available in Ref. [41] for simulation and experimental studies.

Besides the reconstruction speed, the VEM-FPM can also recover the large aberration up to 7π without any prior knowledge of the aberrations. We evaluate the VEM-FPM by means of both simulation and experimental studies as shown in Fig. 5. As given by Figs. 5(a) and 5(b), the VEM-FPM obtained high and stable scores on both PSNR and SSIM compared

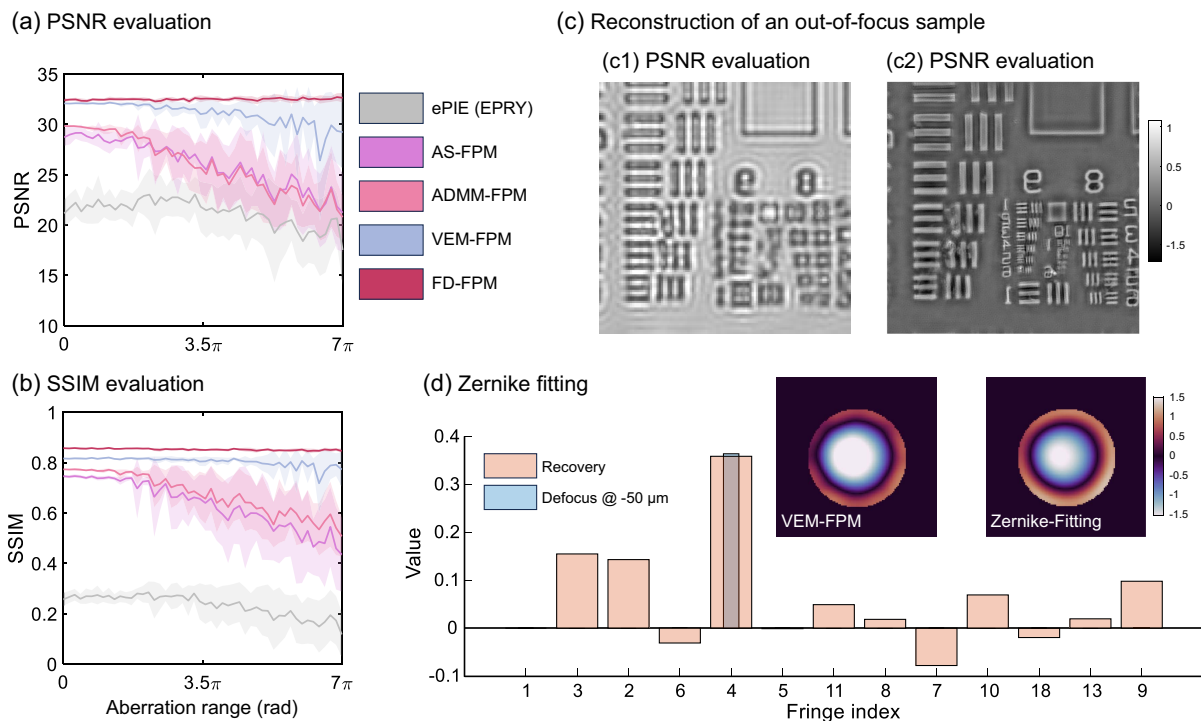


Fig. 5. Validation of VEM-FPM in recovery of aberrations. Simulation studies are performed with 500 groups of testing (a) PSNR and (b) SSIM evaluations for reconstruction quality. (c) Experimental study of VEM-FPM in the blind recovery of the defocused sample. The defocusing distance is known as the ground truth but is not given in the reconstruction. The initial guess of the pupil function is a binary circular mask. (c1) Raw image illuminated by the central LED. (c2) Reconstructed phase pattern. (d) Zernike fitting results for the reconstructed pupil function. The pupil function is smoothed by a Gaussian kernel during the reconstruction.

to conventional FPM algorithms including ePIE (EPRY), ADMM-FPM, and tPIE with adaptive step-size (AS-FPM). The VEM-FPM is also compatible with FD-FPM when the aberration is small. We further perform an experimental study to validate the VEM-FPM, given in Fig. 5(c). The sample is manually defocused at $z = -50\ \mu\text{m}$, as shown in Fig. 5(c1) where the diffraction effect is significant, and the sharp refocused-sample is recovered by the VEM-FPM, as given in Fig. 5(c2). Through Zernike fitting, the aberration can be analyzed at each component of the Zernike polynomial, where our reconstructed result is highly consistent with the ground truth value given by the fringe index No. 4, indicating the presence of high-order aberrations in the imaging system.

5. CONCLUSION

The proposed VEM-FPM in this research is the first model in the field of phase retrieval to utilize variational Bayesian inference. The latent variable $\mathbf{Z}$ links the real-valued intensity detection with the complex-valued wavefront. The physical interpretations of the two steps of VEM-FPM are straightforward. In the E -step, the complex amplitude of $\mathbf{Z}$ is derived from the current estimation of the sample's wavefront and the experimentally detected low-resolution images. This step is akin to the common phase retrieval problem, as commonly applied in inline holography and wavefront shaping. The phase retrieval in the E -step draws benefits from FD-FPM and incorporates feature-domain optimization. Since the optimization operates on low-resolution images, the computation is efficient. Once $\mathbf{Z}$ is determined, the sample wavefront and pupil function are obtained by solving a complex deconvolution problem in the M -step, which is purely quadratic and has a closed-form solution. This deconvolution process is similar to Wiener filtering and certain variations of ADMM [57]. By integrating the plug-and-play technique [58], denoisers can be incorporated into the deconvolution process to further improve the results. Although the VEM-FPM in this research is dedicated to the FPM inverse problem, its computational framework can be extended to other phase retrieval tasks such as classical ptychography, coded ptychography, and inline holography [59]. Perhaps even more importantly, this unique VEM framework can immediately benefit a wide range of applications related to phase retrieval, including biomedical imaging [60] and remote sensing [61–63], and it has a profound impact on biomedicine, optical computing, and wavefront shaping.

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Data Availability. Data and MATLAB codes underlying the results presented in this paper are available in Ref. [64].

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